VIT
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

SCHOOL OF ELECTRONICS ENGINEERING

Course : Antenna and Microwave Engineering Lab (BECE305L)
Lab Slot : L23+L24
Faculty Name : YOGESH KUMAR CHOUKIKER

Task/ Experiment No. : 1	**Date : 5/09/2025**
Name of the Student : HARI KRUTHIK SK	**Regd. No. : 23BEC0358**

TWO PORT NETWORKS AND QUARTER WAVE TRANSFORMER:

CASE 1)

ε = 2.2
H = 1.58 mm
F = 3GHz
t = 0.035 mm
Z0 = 50 ohms
L = 0.02

CASE 2)

ε = 4.4
H = 1.6 mm
F = 4GHz
t = 0.035 mm
Zout = 73 ohms
Zqwt = 60.41 ohms
L = 0.02

AIM :

To study and analyze the performance of different two-port networks by determining their scattering parameters, and to design and investigate a Quarter Wave Transformer for impedance matching between transmission lines of unequal characteristic impedances.

SOFTWARE USED :

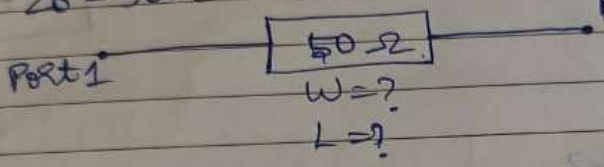
AWR Design Environment (Microwave Office) by **Cadence (formerly by NI/AWR)**

Antenna & Microwave Engg:

1) Lab task:

$\varepsilon_r = 4.4$
$H = 1.6\,mm$
$f = 2.4\,GHz$
$Z_0 = 50\,\Omega$

$S_{12} = S_{21}$
S_{11}, S_{22}



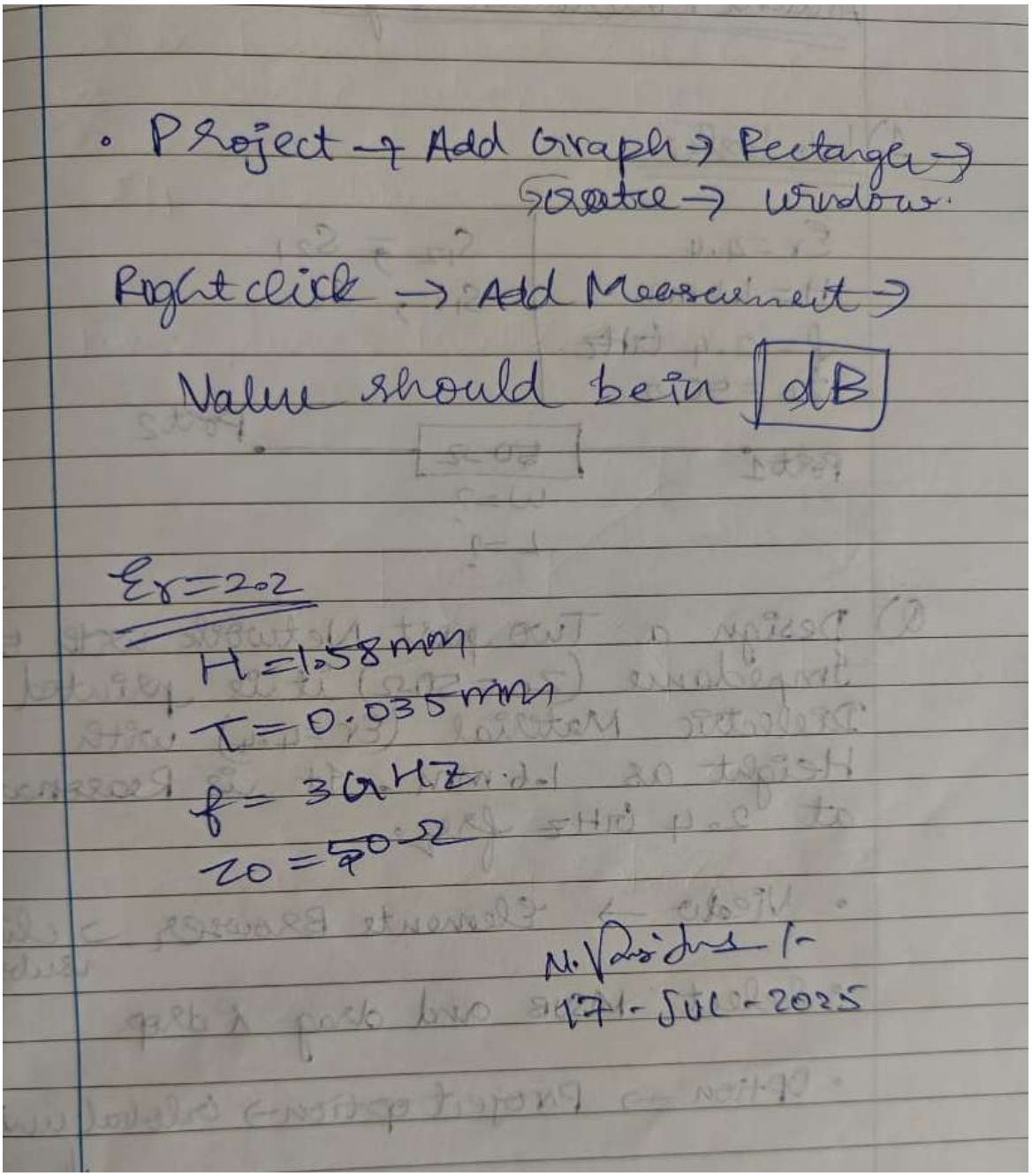
Q) Design a Two port Network with 50Ω Impedance ($Z_0 = 50\Omega$) it is printed on Dielectric Material ($\varepsilon_r = 4.4$) with Height as 1.6 mm. It is Reasonating at 2.4 GHz freq.

- Viedo → Elements Browser → click substrate.
- Select MSUB and drag & drop.
- Option → Project option → Global units ↓ edit units.
- Tool → TX line Calculator

Note the ε_r value → 4.4

Change the conductor to Copper from Silver

Loss Tangent = 0.002


• Project → Add Graph → Rectangle →
Create → Window.
Right click → Add Measurement →
Value should be in dB
Er = 2.2
H = 1.58 mm
T = 0.035 mm
f = 3 GHz
Z0 = 50 Ω
N. Vaishnavi
17-Jul-2025

24/7/25

Q) Task: 9

Design a quarter wave transformer to match a RF transreciever whose source impedence is 50Ω to a dipole antenna which operate at 4GHz. The height of substrate given is FR4. 1.6mm x 4.4 as dielectric cons. Thickness — 0.035mm

Sola

O/P is connected with the Antenna.

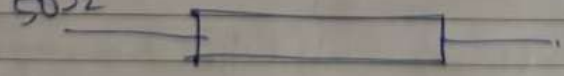
Dipole Anttena Impedance is 73Ω

Length of T.L = λ/4

Resonance = 4GHz.

I/P Impedance = 50Ω.

$$QWT = \sqrt{Z_m \times Z_L}$$



$$QWT = \sqrt{50 \times 73}$$

$$= 60.41522\ \Omega$$

NAVANEETH P
24-Jul-2025
23BEC0358

Expt. No. : 1

Date : 17/07/2025

Two Port Networks

Objective: To study the performance of different two port networks by determining their scattering parameters.

Equipment required: AWR Microwave Office software

2-Port Network

Specifications:

Characteristic impedance	Z_0 = 50Ω
Operating frequency	f = 3GHz
Substrate thickness	H = 1.58 mm
Metal thickness	T = 0.035mm
Dielectricconstant	ε_r = 2.2
Loss tangent	L = 0.02.

Theory :

Microstrip lines: The simple microstrip line uses a single strip conductor on the dielectric that rests on a single ground plane.Generally the ground plane made up of with good conductor like silver or copper and the material used for the dielectric is Teflon or Aluminumor Silicon, etc..It is possible to use several independent strips with the same ground planes and dielectric.Microstrip lines use quasi TEM mode of propagation.The ground plane of the microstrip line must be wide compared with the top conductor, so it appears like a nearly infinite wide ground plane with only very small electric field fringes at its edges.The characteristic impedance of a microstrip line depends on the strip line width, thickness, the distance between microstrip line and ground plane and the dielectric constant of the dielectric material.

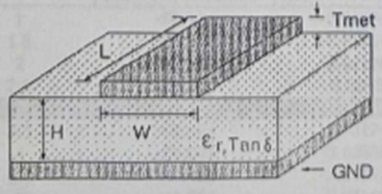


Figure1 (Microstrip Line)

2) To Design a Quarterwave transformer (QWT)

Specifications:-

$L = \lambda/4$

$Z_{out} = 73\Omega$

$T = 0.035\ mm$

$H = 1.6\ mm$

$f = 4GHz$

$Z_{QWT} = 60.41\Omega$

$Z_{QWT} = \sqrt{Z_{in}} \times Z_L$

Design Equations: The effective dielectric constant is calculated by:

$$\varepsilon_e = \frac{\varepsilon_r + 1}{2} + \frac{\varepsilon_r - 1}{2\sqrt{1 + 12\left(\frac{H}{W}\right)}}$$

$$\frac{W}{H} = \begin{cases} \dfrac{8e^A}{e^{2A} - 2} \\ \dfrac{2}{\pi}\left[B - 1 - \ln(2B - 1) + \dfrac{\varepsilon_r - 1}{2\varepsilon_r}\left\{\ln(B - 1) + 0.39 - \dfrac{0.61}{\varepsilon_r}\right\}\right] \end{cases} \text{for} Z_0 > 44 - 2\varepsilon_r \text{narrowstrip}$$

$$A = \frac{Z_0}{60}\sqrt{\frac{\varepsilon_r + 1}{2}} + \frac{\varepsilon_r - 1}{\varepsilon_r + 1}\left(0.23 + \frac{0.11}{\varepsilon_r}\right)$$

$$B = \frac{377\pi}{2Z_0\sqrt{\varepsilon_r}} \text{ forwidestrip}$$

$$\beta l = \phi \qquad l = \frac{\phi}{\beta} \qquad \beta = \frac{2\pi}{\lambda_g} \qquad l = \frac{\phi\lambda_g}{2\pi}$$

W=Width of the microstrip line,l=Length of transmission line,H=Thickness of the substrate, A,B constants,Φ = Phase shift, λ_g=Guide wavelength.

Sample Observations: The behavior of a two port network when matched with50 ohm at both input and output ports for a typical microstrip line with the following specifications is shown below.

Z_0= 50Ω *f*=3GHz, *H*=1.6mm, *T*=0.036 mm, ε_r=4.4, *L*= 0.001

Table: S-Parameters in dB

Frequency (GHz)	DB(\$(1,1)) TWO PORT NETWORK	DB(\$(2,1)) TWO PORT NETWORK
1	-57.116	-0.013305
1.5	-55.299	-0.017447
2	-57.766	-0.021264
2.5	-66.919	-0.02488
3	-125.55	-0.028364
3.5	-62.624	-0.031749
4	-51.042	-0.035085
4.5	-45.453	-0.038414
5	-43.599	-0.041668
5.5	-46.495	-0.044721
6	-58.031	-0.047735

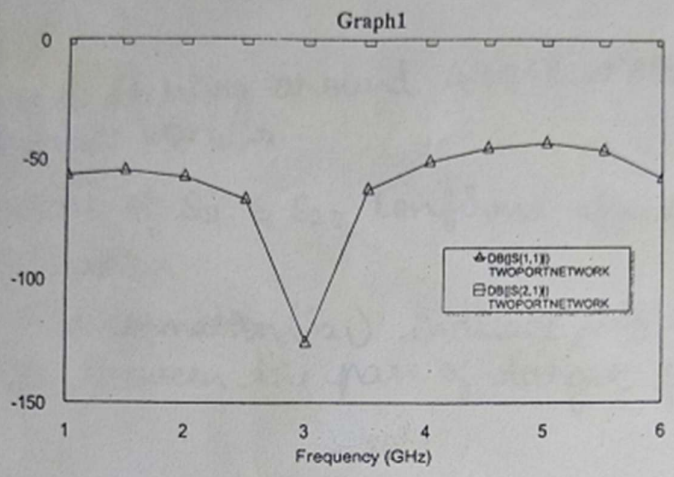
Quarter - Wave Transformer (QWT)

[Tabulation] 2-port Network → f = 4 GHz.

Frequency GHz	S_{11}	S_{12}	S_{21}	S_{22}
3.97	-51.008	-0.00768	-0.00768	-51.008
3.98	-53.699	-0.007681	-0.007681	-53.699
3.99	-57.546	-0.00770	-0.00770	-57.544
4	-63.858	-0.00774	-0.00774	-63.858
4.04	-51.807	-0.00777	-0.00777	-51.807
4.08	-45.042	-0.00783	-0.00783	-45.042
4.1	-42.956	-0.00785	-0.00785	-42.956
4.14	-34.866	-0.00789	-0.00789	-34.866
4.16	-38.654	-0.00831	-0.00831	-38.654
4.18	-37.591	-0.0084	-0.0084	-37.591
4.26	-34.303	-0.00859	-0.00859	-34.303

Model graph:

S-Parameter Plot



Practical Observations:

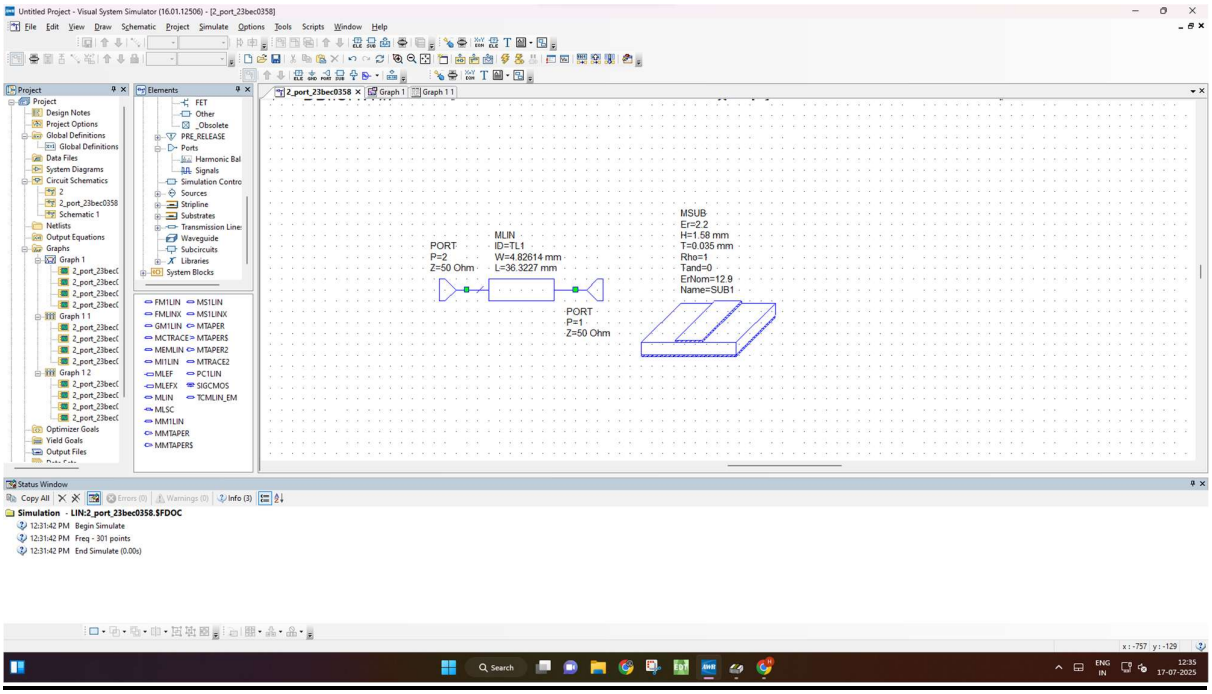
2-Port Network $f = 3GHz$.

Frequency Mhz/ghz	S_{11}	S_{12}	S_{21}	S_{22}
2.90	-84.863	-0.014434	-0.014434	-84.863
2.92	-86.855	-0.014403	-0.014403	-86.855
2.94	-89.392	-0.014428	-0.0144	-89.392
2.96	-92.939	-0.014449	-0.014449	-92.939
2.98	-98.985	-0.0452	-0.0452	-98.985
3	-120.7	-0.014598	-0.014598	-120.7
3.02	-98.664	-0.01460	-0.01460	-98.664
3.04	-92.642	-0.014467	-0.014467	-92.642
3.06	-89.043	-0.014696	-0.014696	-89.043
3.08	-86.434	-0.014718	-0.014718	-86.434
3.09	-85.346	-0.01481	-0.01481	-85.346

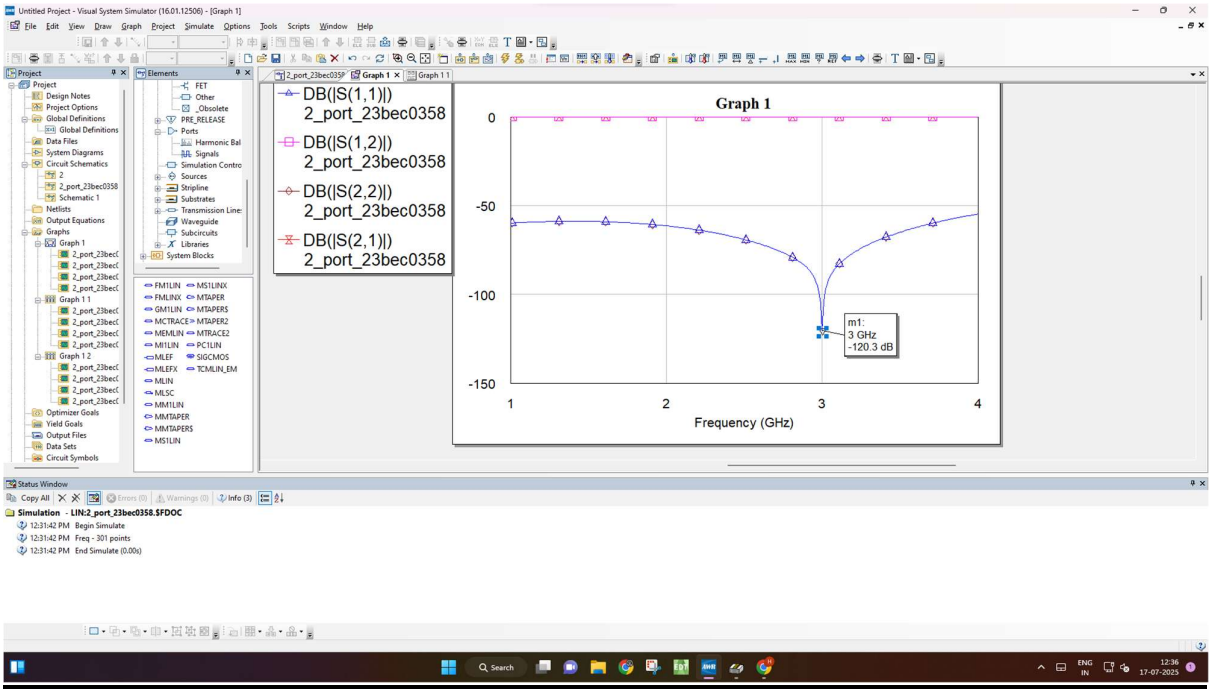
Conclusions:

- Improves matching around 4GHz with minimizes reflection at 60.41Ω.
- Low Values of S_{11} & S_{22} confirms effective impedance Transformation.
- High Transformation (S_{21}) indicate efficient energy transfer between the pass of target frequency.

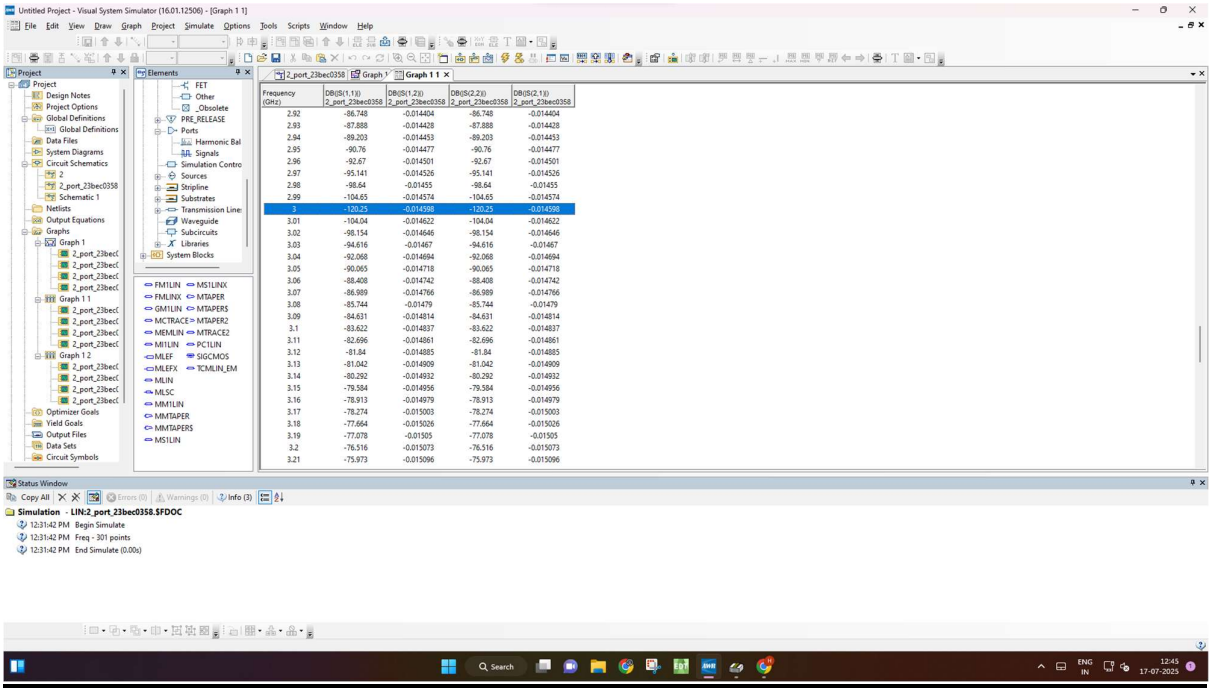
SCHEMATIC DIAGRAM : [TWO PORT]



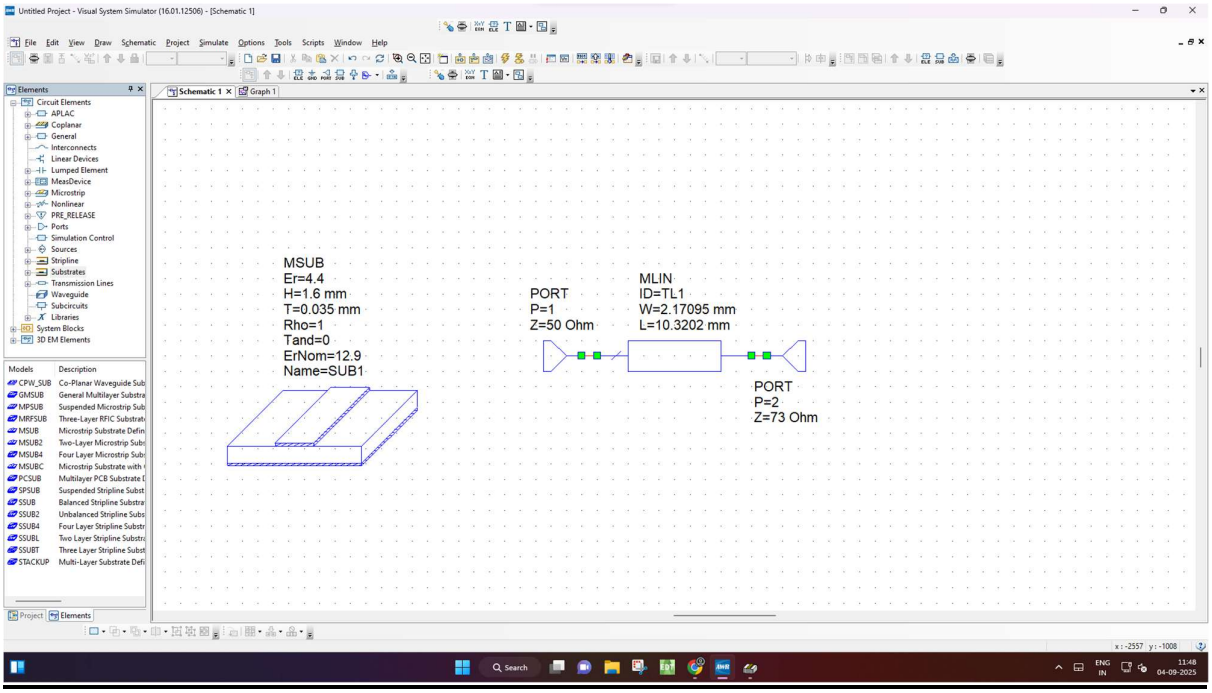
GRAPH :



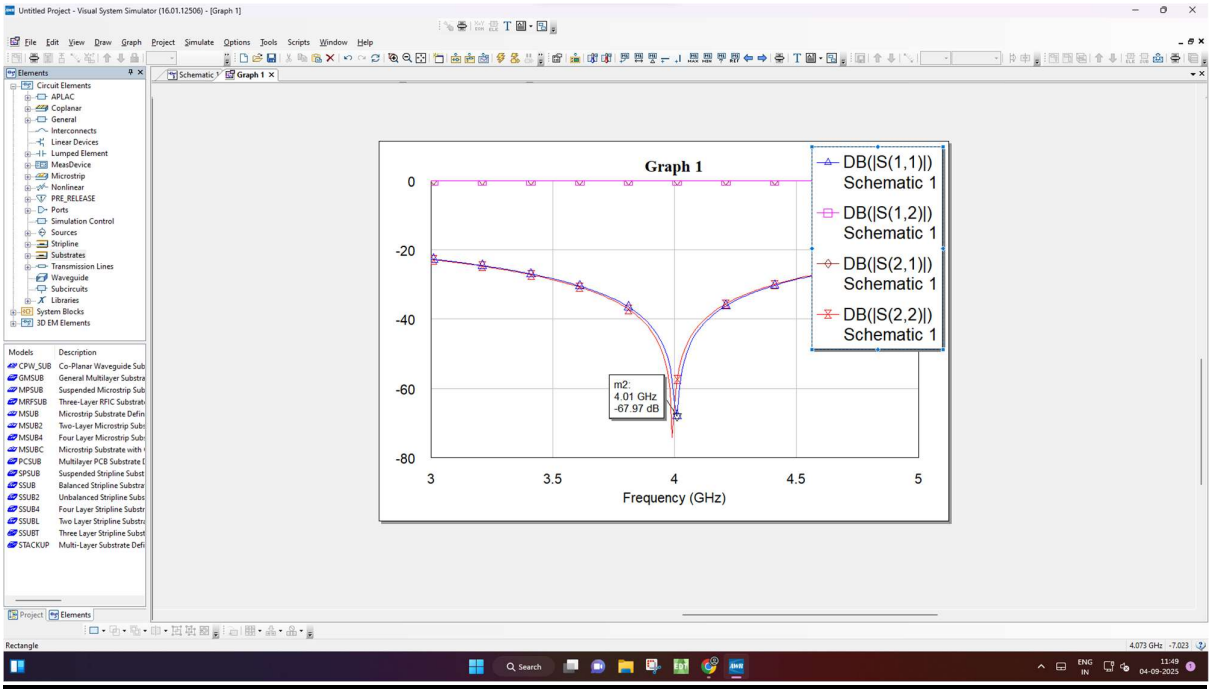
TABULAR COLUMN :



SCHEMATIC DIAGRAM : [QWT]



GRAPH :



TABULAR COLUMN :

